

Dillon Blyth

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Profile

BSc Computer Science student at the University of Dundee with a strong foundation in data structures, algorithms, and OOP (C++/Java, Python). Passionate about full-stack software engineering and AI model efficiency (post-training quantization, compiler pass design). Seeking a Software Engineering or AI/Systems Year in Industry/Summer Internship to apply problem-solving and system optimization skills to scalable, real-world products.

Technical Skills

- **Languages:** C++, Python, Java, C, JavaScript, SQL, HTML5, CSS3
- **Frameworks & Web:** PyTorch, React Native, Node.js, Express.js, Bootstrap
- **Tools & Databases:** MongoDB, MySQL, SQLite, Git/GitHub, VS Code, Unix/Linux, CI/CD
- **Core Domains:** Data Structures & Algorithms, AI Model Quantization (INT8/PTQ), MLIR, Graph Search

Technical Projects

Badminton Training Mobile Application | *React Native, Python, SQLite*

- Architected a full-stack, cross-platform mobile app featuring interactive footwork tutorials and user session management.
- Designed an SQLite database schema for tracking user progression metrics and personalized profiles.
- Integrated device hardware APIs for on-device video capture and local media storage to review drill performance.
- Implemented algorithmic movement patterns to dynamically guide users through training drills step-by-step.

Pathfinding Optimisation AI | *Python, Graph Theory*

- Built an AI pathfinding system using weighted graphs and priority queues to reduce wait times in high-density environments.
- Applied Dijkstra and A* heuristic cost functions, benchmarking algorithmic complexity against runtime-optimality trade-offs.

Object-Oriented Software Suite | C++, Java

- Engineered modular data structures (binary search trees, custom arrays) for search and inventory management systems, utilizing rigorous unit testing and memory management principles.

Education

BSc (Hons) Computer Science | University of Dundee *September 2024 – June 2028 (Expected)*

- **Key Coursework:** Data Structures & Algorithms, Systems Architecture, Software Engineering, Web Development.

Experience & Leadership

Badminton Club Convenor & Team Player | University of Dundee *September 2024 – Present*

- Manage university-wide tournaments and dynamic ladder competitions, organizing match scheduling and court allocations.
- Liaise with sports union officials to coordinate logistics, demonstrating strong communication and event management skills.

Chef | Pacamara Food and Drink & ESS Compass *July 2025 – Present*

- Balance 20 hours of weekly employment alongside full-time Computer Science studies, demonstrating strong self-discipline.
- Execute high-volume food preparation under strict time constraints, averaging 30+ meals per hour during peak service.

Peer Tutor (Computer Science, Maths, Chemistry, Biology and Physics) | Woodmill High School *August 2022 – June 2024*

- Mentored junior students by breaking down complex algorithmic, logical, and mathematical concepts into accessible steps.

References available upon request.